DOMiNANCE CVS MOCK EXAMINATION 02

Cardiovascular Physiology | 60 questions | 90 minutes | 60 marks

Choose the single best answer (A-D) for each question. The answer key is on the page immediately after the last question.

1. How does normal inspiration tend to assist systemic venous return?

A. It closes all systemic venous valves simultaneously.
B. It increases peripheral venous capacitance until return stops.
C. Lower chest pressure draws venous blood toward the heart.
D. It increases right atrial pressure and reduces abdominal pressure.

2. Arterial pressure inside an arteriole is unchanged, but pressure in its surrounding tissue rises. What happens to transmural pressure?

A. It increases because the pressure difference widens.
B. It decreases because the pressure difference narrows.
C. It becomes identical to intravascular arterial pressure.
D. It remains fixed because arterial pressure is unchanged.

3. Why does the fetal cardiovascular system divert much blood away from the pulmonary circuit before birth?

A. Pulmonary vascular resistance is exceptionally low before birth.
B. Pulmonary arterial pressure stays below placental venous pressure throughout fetal life.
C. The placenta oxygenates blood; fetal lungs resist flow.
D. Umbilical arteries carry already oxygenated blood to the heart.

4. A patient develops cardiac tamponade. Why can stroke volume fall despite a normal blood volume?

A. High pericardial pressure restricts ventricular filling.
B. Tamponade increases functional ventricular compliance.
C. Tamponade directly raises venous return above normal.
D. The higher pericardial pressure directly increases end-diastolic ventricular volume.

5. Heart rate rises from 70 to 80 beats/min while stroke volume stays at 70 mL. By how much does cardiac output increase?

A. By 1.40 litres per minute.
B. By 0.70 litres per minute.
C. By about 0.49 litres per minute.
D. By 0.80 litres per minute.

6. A burn patient presents with shock and an initially high haematocrit. What explains the contrast with recent whole-blood haemorrhage?

A. Burns cause immediate erythropoietin-driven erythrocytosis.
B. Plasma leakage leaves intravascular red cells concentrated.
C. Burns produce increased circulating volume despite shock.
D. Burns selectively destroy red cells circulating in blood.

7. During graded exercise, which combination helps increase cardiac output?

A. Slower pacemaking and lower venous return.
B. Reduced sympathetic drive and greater afterload.
C. Higher rate, contractility and venous return.
D. Greater vagal activity and smaller stroke volume.

8. A patient has low circulating volume. Which hormone chiefly increases renal water retention and can constrict vessels at high concentration?

A. Endothelial prostacyclin.
B. Vasopressin, also called ADH.
C. Vasoactive intestinal peptide.
D. Atrial natriuretic peptide.

9. How does local hypoxia affect most systemic arterioles versus pulmonary arterioles?

A. Both primarily constrict to lower oxygen delivery.
B. Both remain relatively insensitive to tissue oxygen.
C. Systemic vessels dilate; pulmonary vessels constrict.
D. Systemic vessels constrict while pulmonary vessels dilate.

10. After a meal containing long-chain fats, which pathway carries these lipids from the gut?

A. Renal filtration of absorbed fats.
B. Direct red-cell delivery to lacteals.
C. Oxygenated blood flow through the portal vein alone.
D. Chylomicron transport from intestinal lacteals.

11. After three minutes upright, pressure changes from 126/82 to 104/77 mmHg. How is this classified using the chapter criterion?

A. No orthostatic hypotension because diastolic pressure fell only 5 mmHg.
B. Orthostatic hypotension only if both pressures fall by 20 mmHg.
C. Orthostatic hypotension because systolic pressure fell 22 mmHg.
D. No orthostatic hypotension because final systolic pressure exceeds 100 mmHg.

12. Cardiac output falls by 20%, but total peripheral resistance rises by 25%. What happens to mean arterial pressure in the simplified MAP relation?

A. It decreases by approximately one quarter.
B. It increases by approximately one quarter.
C. It decreases by approximately one fifth.
D. It remains approximately unchanged.

13. Why might increasing heart rate far above normal eventually reduce cardiac output?

A. More beats per minute proportionally lengthen each ejection phase.
B. Diastolic time becomes longer and increases filling.
C. Systemic afterload falls enough to prevent forward ventricular ejection.
D. Shorter diastole reduces filling and stroke volume.

14. Why is a patient with significant coronary narrowing more vulnerable to angina when heart rate rises sharply?

A. The coronary sinus develops arterial hypertension.
B. The P wave disappears and blocks blood flow.
C. Shorter diastole and higher myocardial demand.
D. Myocardial oxygen consumption falls below baseline.

15. A pressure tracing shows atrial contraction shortly after an ECG P wave and ventricular contraction shortly after QRS. What principle explains both delays?

A. The T wave must occur before either chamber can begin contraction.
B. Electrical excitation normally precedes the mechanical response of cardiac muscle.
C. Atrial and ventricular pressures directly create the corresponding ECG waves.
D. Mechanical contraction normally finishes before electrical activation begins in the chamber.

16. Which statement best reflects the updated interpretation of capillary fluid balance?

A. Interstitial oncotic forces remain independent of the endothelial glycocalyx.
B. Most capillaries continuously reabsorb nearly all filtered fluid at their venous ends.
C. The endothelial surface layer has little effect on filtration at rest.
D. The glycocalyx regulates filtration; lymph returns excess fluid.

17. A patient with left-sided forward heart failure reports fatigue and produces very little urine. Which mechanism links the two?

A. Reduced systemic perfusion limits muscle and renal blood flow.
B. Elevated venous pressure increases renal filtration.
C. Pulmonary congestion causes systemic arteriolar dilation.
D. Increased left ventricular output stimulates renal water loss.

18. Once local metabolic dilation restores an organ's blood flow, what prevents dilation from continuing unchecked?

A. The restored flow clears accumulated dilator metabolites.
B. The restored flow removes oxygen from the tissue cells.
C. The restored flow causes a sustained rise in local lactate.
D. The restored flow abolishes vessel-wall smooth muscle.

19. A patient with marked polycythaemia has elevated haematocrit. What is the expected direct effect on systemic flow resistance?

A. Higher viscosity and lower resistance.
B. Lower viscosity and higher resistance.
C. Higher viscosity and higher resistance.
D. Lower viscosity and lower total vascular resistance.

20. The ST segment is nearly flat despite an actively contracting ventricle. What electrical state best explains that appearance?

A. The ventricle has completed repolarisation and returns to electrical rest.
B. The atria and ventricles are depolarising in opposite directions.
C. Most ventricular cells are depolarised together in their plateau phase.
D. The AV node is holding all impulses before ventricular activation.

21. Which pressure relationship helps prevent fluid accumulation in normal pulmonary alveoli?

A. Absence of plasma proteins in pulmonary capillaries.
B. Higher alveolar protein pressure than in capillary plasma.
C. Pulmonary capillary pressure consistently exceeds systemic arterial pressure.
D. Relatively low pulmonary capillary hydrostatic pressure.

22. Increasing capillary hydrostatic pressure while other Starling forces stay unchanged tends to produce what?

A. Decreased outward fluid filtration.
B. More fluid filtration into tissues.
C. Increased plasma oncotic attraction.
D. Reversal of lymphatic flow.

23. Which contrast best distinguishes forward failure from backward failure?

A. Forward indicates excess cardiac output; backward low afterload.
B. Forward is confined to the right heart; backward to the left.
C. Forward is venous congestion; backward is low organ blood flow.
D. Forward reduces delivery; backward causes venous congestion.

24. A region of heart muscle starts working harder. Which local product of its own metabolism helps increase its blood flow?

A. Adenosine formed as ATP is used up.
B. Noradrenaline from sympathetic nerves.
C. Angiotensin II formed in the blood.
D. Serotonin released by nearby platelets.

25. Which pressure difference primarily drives blood back to the right atrium from the body?

A. Aortic systolic minus right ventricular systolic pressure.
B. Pulmonary venous minus pulmonary capillary pressure.
C. Left ventricular systolic minus aortic diastolic pressure.
D. Systemic filling pressure minus right atrial pressure.

26. A severely underperfused patient begins breathing rapidly, although the lungs were initially healthy. What drives this change?

A. Increased aerobic ATP production stimulates breathing.
B. Suppression of the peripheral chemoreceptors stimulates breathing.
C. Lactic acidosis stimulates peripheral chemoreceptors.
D. High tissue oxygen content increases respiratory drive.

27. The CSF volume is about 150 mL and production is 550 mL/day. Approximately how many complete turnovers occur per day?

A. About 2.
B. About 1.
C. About 4.
D. About 6.

28. What does the mechanical stroke work of a ventricle represent?

A. Electrical energy in the SA node over one minute.
B. The number of coronary capillaries recruited.
C. Heart rate multiplied by body surface area.
D. Ventricular pressure-volume work per beat.

29. A technician places V1 at the fourth intercostal space. Where should the electrode be positioned horizontally?

A. At the anterior axillary line.
B. At the right sternal border.
C. At the left midaxillary line.
D. At the left midclavicular line.

30. A bleeding patient has a nearly normal arterial pressure but cold skin and reduced urine output. What is the best conclusion?

A. Normal pressure excludes significant shock.
B. Low perfusion can coexist with normal pressure.
C. Blood flow to all organs must remain normal.
D. Cardiac output must be above its pre-bleeding level.

31. Why may advanced liver disease cause generalised oedema even without venous obstruction?

A. Less albumin synthesis reduces plasma oncotic pressure.
B. Liver failure directly shuts every lymphatic valve.
C. Excess albumin synthesis draws water into tissues.
D. A rise in plasma protein concentration stops absorption.

32. At one instant, Lead I records +0.4 mV and Lead III records -0.2 mV. What must Lead II record?

A. -0.2 mV, from using the third lead alone.
B. +0.6 mV, from adding their magnitudes.
C. -0.6 mV, from subtracting the two values.
D. +0.2 mV, from adding the signed voltages.

33. An exercising leg receives increased blood flow while its muscle continues contracting. What is this response called?

A. Active hyperaemia driven by ongoing tissue metabolism.
B. Autoregulation that holds its flow at rest level.
C. Baroreflex vasodilation after systemic arterial pressure rises.
D. Reactive hyperaemia following a temporary flow block.

34. A harsh systolic murmur develops during blood ejection through a narrowed aortic valve. What mechanism explains it?

A. Opening of the aortic valve during ventricular relaxation.
B. Laminar blood flow through a normal-sized valve.
C. Regurgitation through the mitral valve during diastole.
D. Turbulent forward flow through a stenotic outflow orifice.

35. A highly active cortical region accumulates carbon dioxide and hydrogen ions. What is the expected local vascular response?

A. Local arterioles dilate, increasing perfusion.
B. Venules constrict without an arteriolar response.
C. Local arterioles constrict, reducing perfusion.
D. Capillaries close while arteriolar tone stays fixed.

36. Lead II has an upright QRS while aVR has a downward QRS during the same beat. What best explains the difference?

A. Atrial and ventricular activity alternate between the leads on each beat.
B. One lead measures depolarisation and the other repolarisation.
C. Each lead projects the same electrical vector onto a different axis.
D. The two leads record different mechanical contractions.

37. In a normal cardiac cycle, how does the rate of ventricular ejection change after the semilunar valves open?

A. Ejection peaks only after ventricular pressure has begun to fall.
B. It starts rapidly and then becomes reduced ejection.
C. It remains identical in flow throughout systole.
D. It starts slowly and accelerates until valve closure.

38. Minutes after a severe allergic exposure, a patient's pressure drops with generalised swelling. Which vascular action is most relevant?

A. Generalised constriction with reduced capillary permeability.
B. Histamine-related dilatation with increased capillary permeability.
C. Selective haemorrhage into the gastrointestinal tract.
D. Increased myocardial pump function with reduced venous capacitance.

39. Why can chronic widespread alveolar hypoxia lead to pulmonary hypertension?

A. Persistent hypoxic constriction increases lung resistance.
B. Persistent hypoxia opens every pulmonary capillary without resistance.
C. Widespread hypoxia lowers pulmonary vascular smooth-muscle tone.
D. Chronic hypoxia eliminates blood passing through the lungs.

40. The jugular venous 'c' wave is chiefly linked to which event?

A. Right atrial contraction immediately before QRS.
B. Systolic bulging of the tricuspid valve.
C. Passive systemic venous filling late in systole.
D. Rapid emptying of the right atrium.

41. An isolated SA node fires at about 100 beats/min, whereas the same person's resting pulse was about 70. Which interpretation best fits?

A. Resting sympathetic activity had restrained SA nodal firing.
B. Atrial conduction had slowed the spontaneous SA nodal rate.
C. Resting vagal activity had restrained SA nodal firing.
D. The cardiac pacemaker was intrinsically limited to 70.

42. A patient with massive haemorrhage remains severely hypotensive after blood volume is completely restored. Which stage fits?

A. Postural hypotension from remaining upright.
B. Early compensated hypovolaemic shock with intact reflexes.
C. Transient vasovagal faint without major tissue damage.
D. Irreversible shock after extensive tissue injury.

43. At an otherwise unchanged capillary, how would increased interstitial protein concentration affect fluid movement?

A. It strengthens plasma oncotic pressure, reducing filtration.
B. It increases protein attraction into the blood compartment.
C. It lowers capillary hydrostatic pressure, opposing filtration.
D. It strengthens the outward oncotic force favouring filtration.

44. Regular P waves continue at one rate and QRS complexes at a slower independent rate, with no fixed PR relationship. Which rhythm disturbance is present?

A. Third-degree atrioventricular block.
B. Sinus rhythm with a normal PR interval.
C. First-degree atrioventricular block.
D. Second-degree atrioventricular block.

45. Two cylindrical vessels have equal transmural pressure, but vessel X has twice the radius of vessel Y. How do their required wall tensions compare?

A. Vessel X requires four times the tension of Y.
B. Both vessels require exactly the same wall tension.
C. Vessel X requires twice the wall tension of Y.
D. Vessel X requires half the wall tension of Y.

46. What physiological purpose is served by slow conduction through the AV node?

A. It allows atrial contraction to finish ventricular filling.
B. It prevents the impulse from ever reaching the ventricular myocardium.
C. It ensures ventricles eject before the atria contract.
D. It allows the aortic valve to remain open during atrial systole.

47. A patient in compensated haemorrhagic shock has cold, pale skin. Which mechanism accounts for this?

A. Increased sympathetic cutaneous vasoconstriction.
B. Loss of vasomotor drive causing widespread dilation.
C. Increased skin capillary blood volume from vasodilation.
D. Direct activation of warm-sensitive skin receptors.

48. A graph of organ flow against perfusion pressure is almost flat over a middle range but slopes downward at very low pressures. Why?

A. The veins become the principal resistance at low pressure.
B. The arterioles reach their maximal dilation at low pressure.
C. The heart stops driving any pressure into the organ.
D. The arterioles reach their maximal constriction at low pressure.

49. A ventricular action potential has a prolonged plateau. Which membrane currents chiefly sustain it?

A. Absence of calcium entry with continued chloride influx.
B. Calcium inflow balanced by outward potassium current.
C. Rapid sodium entry alone throughout the entire plateau.
D. Increased potassium efflux without any inward current.

50. During progressive shock, capillary hydrostatic pressure paradoxically rises while systemic arterial pressure is falling. Which microvascular pattern best explains it?

A. Open capillary entry with constricted venules.
B. Increased venular drainage with reduced capillary entry.
C. Precapillary constriction with freely draining venules.
D. Precapillary constriction with widened venous outlets.

51. During severe hepatic hypoperfusion, cells near the central vein are especially vulnerable to hypoxia. Why?

A. They are closest to arterial inflow.
B. They are located outside the hepatic sinusoidal network.
C. They receive blood with the highest oxygen content.
D. They are farthest downstream from the portal triads.

52. How does calcium entering through L-type channels activate a much larger ventricular calcium signal?

A. It triggers calcium release from the sarcoplasmic reticulum.
B. It closes ryanodine receptors in the sarcoplasmic reticulum.
C. It causes SERCA to release its stored calcium outward.
D. It directly hydrolyses ATP within actin.

53. What extra perspective do the chest leads V1-V6 add to the standard limb-lead views?

A. They replace voltage measurement with intracardiac pressure.
B. They isolate atrial electrical activity from ventricular activity.
C. They remove the need to record the frontal electrical plane.
D. They show cardiac electrical vectors in the horizontal plane.

54. A haemorrhage suddenly lowers blood pressure. Which changes help restore it through the arterial baroreflex?

A. Increased vagal drive and reduced venous return.
B. Reduced sympathetic drive and increased vagal drive.
C. Increased sympathetic drive and reduced vagal drive.
D. Reduced sympathetic drive and greater venous storage.

55. For steady laminar flow through a straight tube, where is linear velocity highest?

A. At the endothelial wall.
B. At the vessel centre.
C. Uniformly across the lumen.
D. At the outer plasma layer.

56. An ECG is repeated at 50 mm/s instead of 25 mm/s, with no change in the patient's conduction. How does the same 0.16-s PR interval appear?

A. It occupies twice the former number of horizontal squares.
B. It occupies the same number of horizontal squares.
C. It occupies half as many horizontal squares as on the slower recording.
D. It occupies twice the former number of vertical squares.

57. An examination reveals raised JVP, a tender enlarged liver, ascites and ankle oedema. Which haemodynamic change unifies these?

A. High systemic venous pressure behind the right ventricle.
B. Increased pulmonary venous pressure without systemic congestion.
C. Reduced filtration from peripheral capillaries.
D. Reduced central venous pressure from depleted blood volume.

58. What produces an S4 in a patient with a stiff ventricle?

A. Turbulence during systolic outflow across a normal aortic valve.
B. Opening of the pulmonary valve against high pressure.
C. Atrial contraction into a poorly compliant ventricle.
D. Rapid early-diastolic filling before any atrial activity.

59. Brain glucose uptake remains adequate despite an intact blood-brain barrier that restricts passage between endothelial cells. What transport process explains this?

A. Through insulin-triggered endothelial pores.
B. Through a specific endothelial transporter.
C. By free movement through tight junctions.
D. Through bulk passage of plasma proteins.

60. A cuff that is far too small is used on a large upper arm. Which measurement error is likely?

A. The cuff underestimates systolic pressure but overestimates diastolic pressure.
B. No systematic effect on cuff pressure.
C. Falsely low BP from excessive arterial unloading.
D. An undersized cuff gives a falsely high reading.

ANSWER KEY

Mock 02 - 60 Questions | One mark each

Check your answers only after completing the examination.

Q01-Q15		Q16-Q30		Q31-Q45		Q46-Q60	
01.	C	16.	D	31.	A	46.	A
02.	B	17.	A	32.	D	47.	A
03.	C	18.	A	33.	A	48.	B
04.	A	19.	C	34.	D	49.	B
05.	B	20.	C	35.	A	50.	A
06.	B	21.	D	36.	C	51.	D
07.	C	22.	B	37.	B	52.	A
08.	B	23.	D	38.	B	53.	D
09.	C	24.	A	39.	A	54.	C
10.	D	25.	D	40.	B	55.	B
11.	C	26.	C	41.	C	56.	A
12.	D	27.	C	42.	D	57.	A
13.	D	28.	D	43.	D	58.	C
14.	C	29.	B	44.	A	59.	B
15.	B	30.	B	45.	C	60.	D

Answer key ends here. Review explanations separately in the question-bank archive.